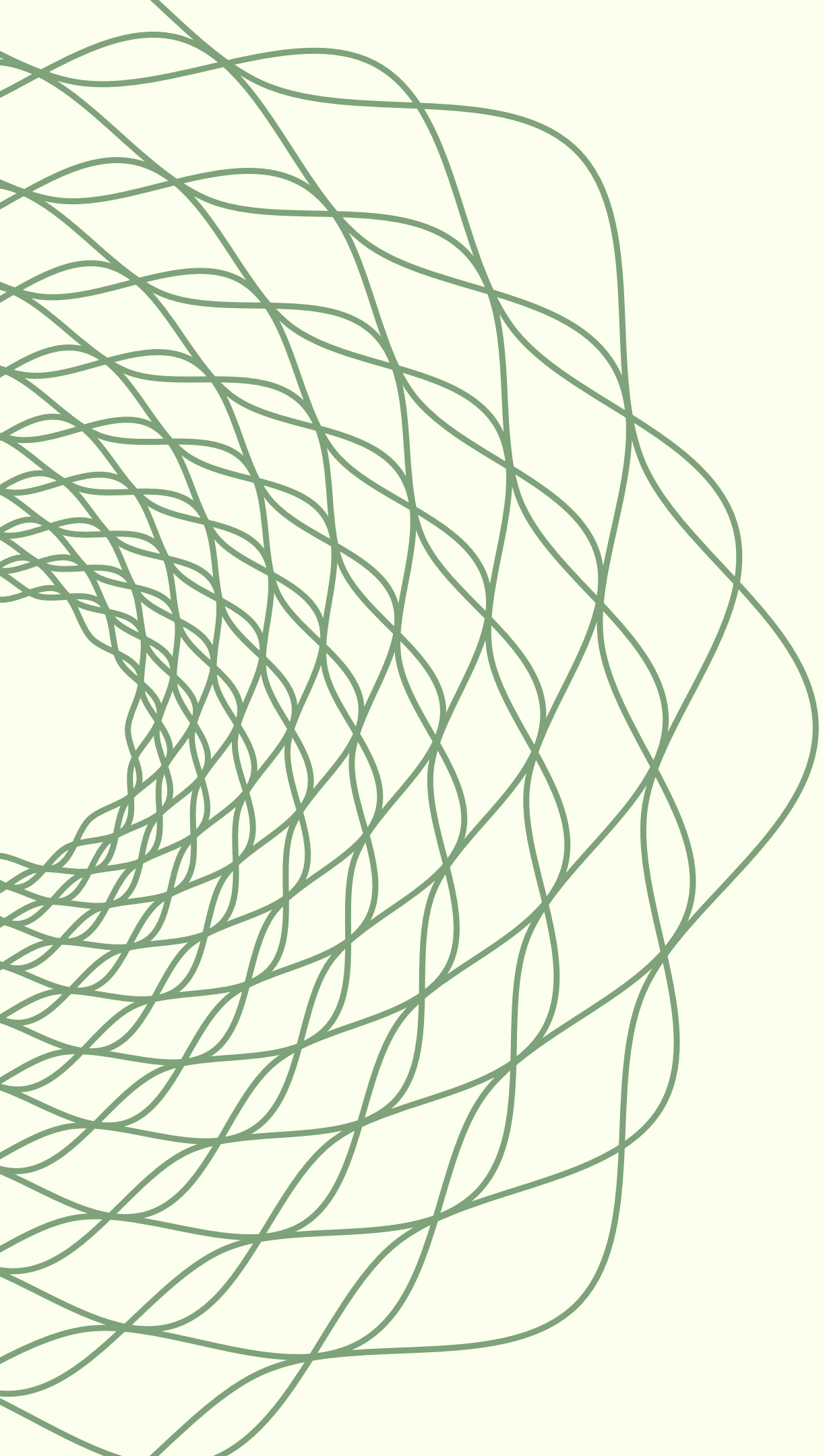




Team: Anal-yst

Trilytics'25

Predicting Model for Farmers
Income

Team Members:

Saksham Sureka

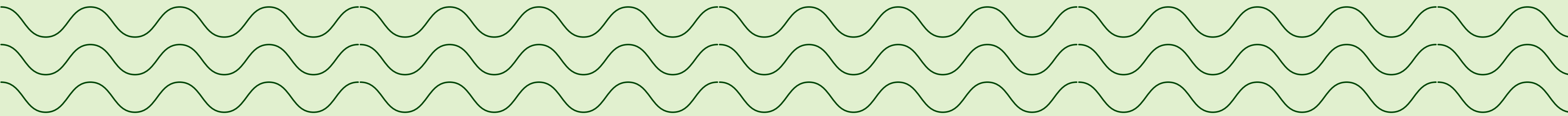
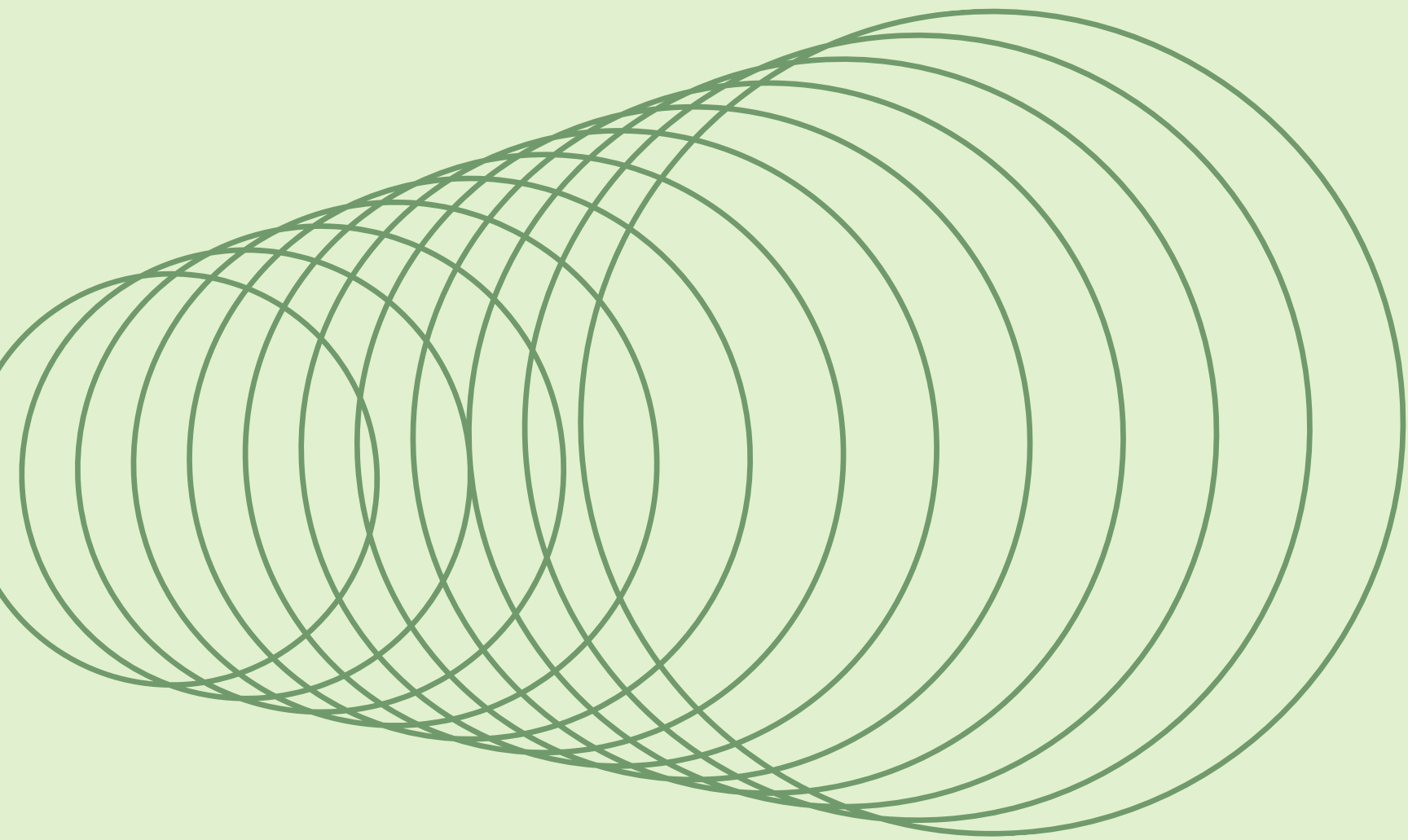
Arjun Kumar

Shreya Waghmode

Manya Singhal

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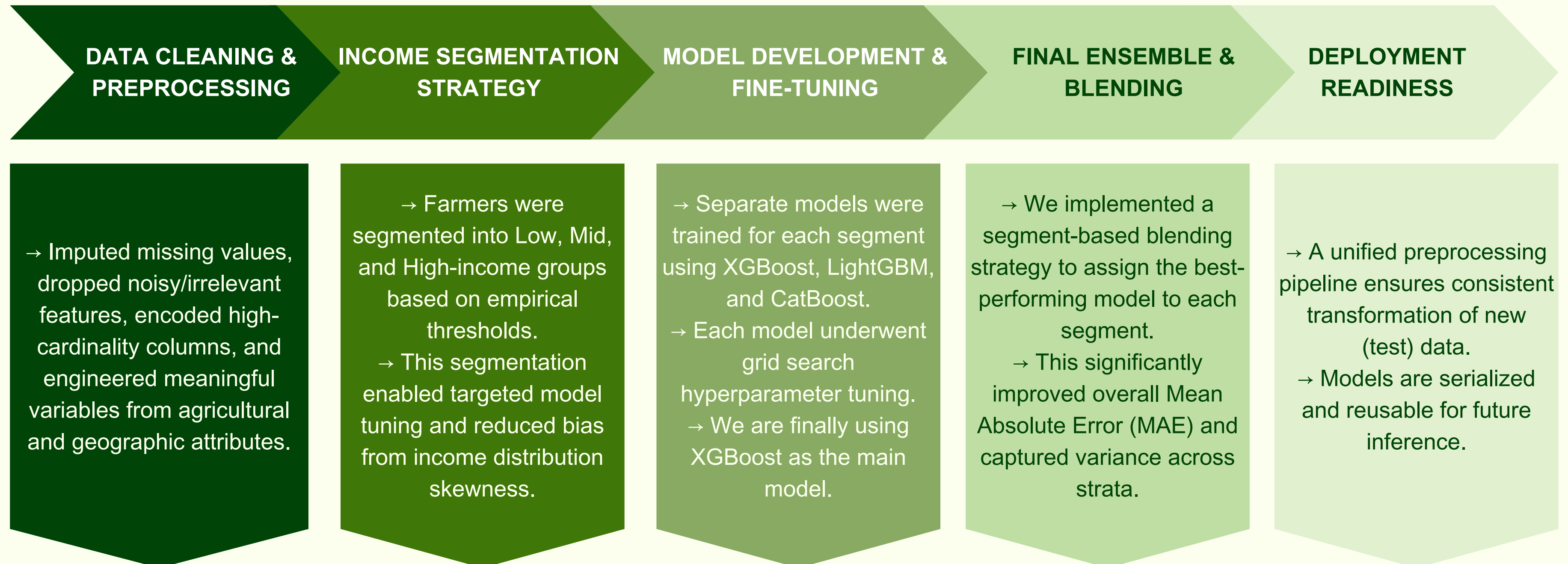
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Executive Summary.

Farmer Income Prediction Model – Strategy & Outcomes

Over the course of this project, we developed a robust and segmented machine learning framework to accurately predict individual farmer incomes across India. Our end-to-end approach integrates domain-specific feature engineering, targeted modeling strategies, and rigorous evaluation, tailored to maximize predictive accuracy while ensuring scalability.



Introduction

PROBLEM STATEMENT

Despite several welfare schemes, Indian farmers still face unstable incomes due to factors like weather, land access, irrigation, market distance, and socio-economic challenges. Accurate income prediction can enable timely support and better resource allocation.

Factors Influencing The Relationship



PROJECT OBJECTIVE

To develop a machine learning model that predicts the total annual income of farmers using a rich set of demographic, agricultural, environmental, and geographical features.

WHAT WE DID

- Cleaned and preprocessed over 70 diverse features from farmer profiles
- Handled missing values and standardized various inputs
- Applied Label and Target Encoding for categorical variables
- Engineered insightful features like regional income trends, land-to-loan ratios, rainfall averages, and market accessibility scores
- Evaluated model performance using MAE and cross-validation
- Built regression models starting with Linear Regression, advancing to XGBoost

Data at a Glance: What Information Did We Use?

RAW DATA

PREPROCESSING

FEATURE
ENGINEERING

MODEL

INCOME
PREDICTION

DEMOGRAPHICS

Gender, Marital Status, Address Type, Land Ownership

LAND AND AGRICULTURE

Total Agri Land, Net Agri Area (% & ha), Season-wise Agri Scores

ENVIRONMENT

Rainfall (mm), Min/Max Temp (Kharif & Rabi, multi-year)

OUR TARGET VARIABLE

Total Income (INR) – The crucial metric we aim to predict.

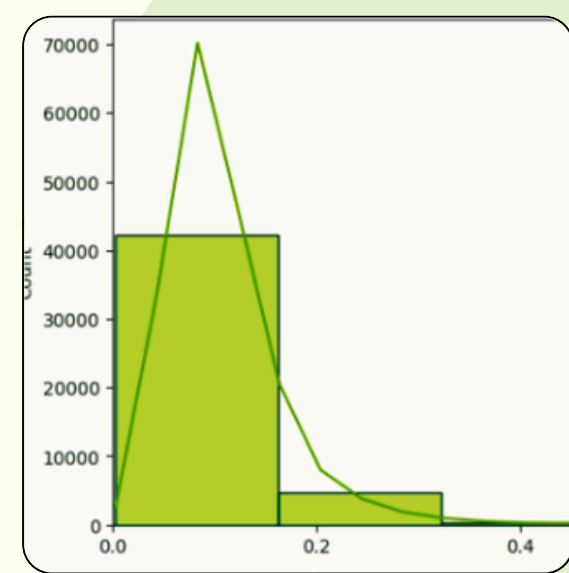
FINANCE AND ACCESS

Avg Loan, Non-Agri Income, Distance to Mandi & Railway

SOCIO-ECONOMIC

Village Type, Socio-Eco Score, Sanitation, Electricity, Housing

Uncovering Patterns in the Data

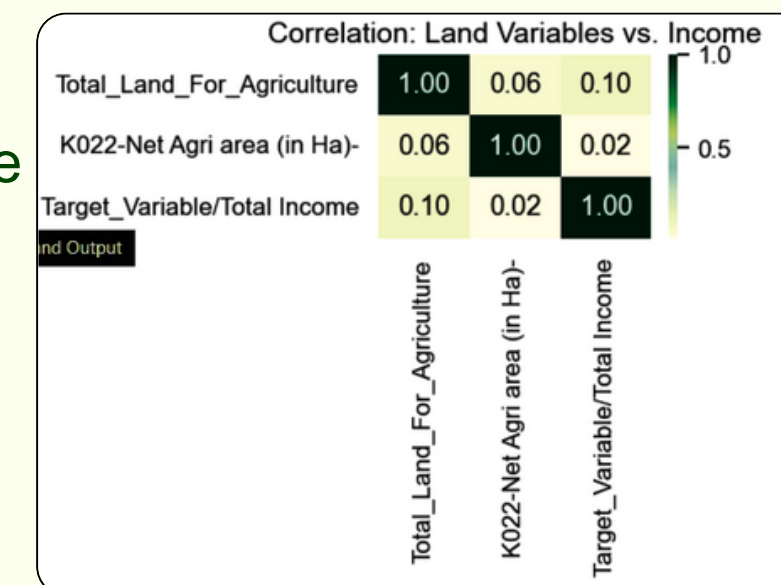
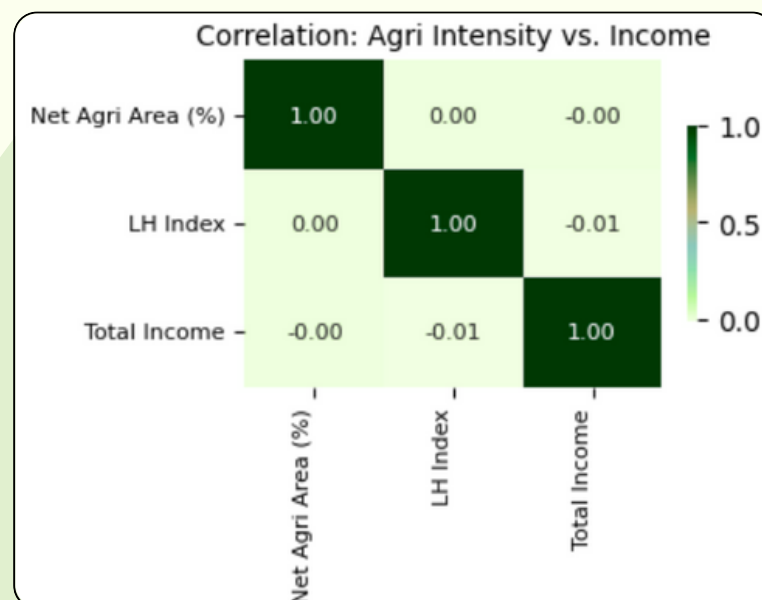
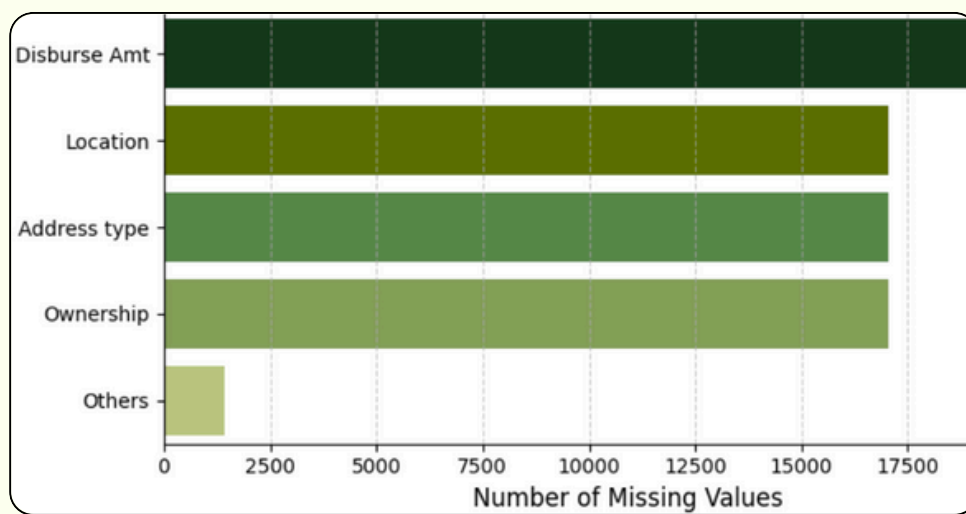


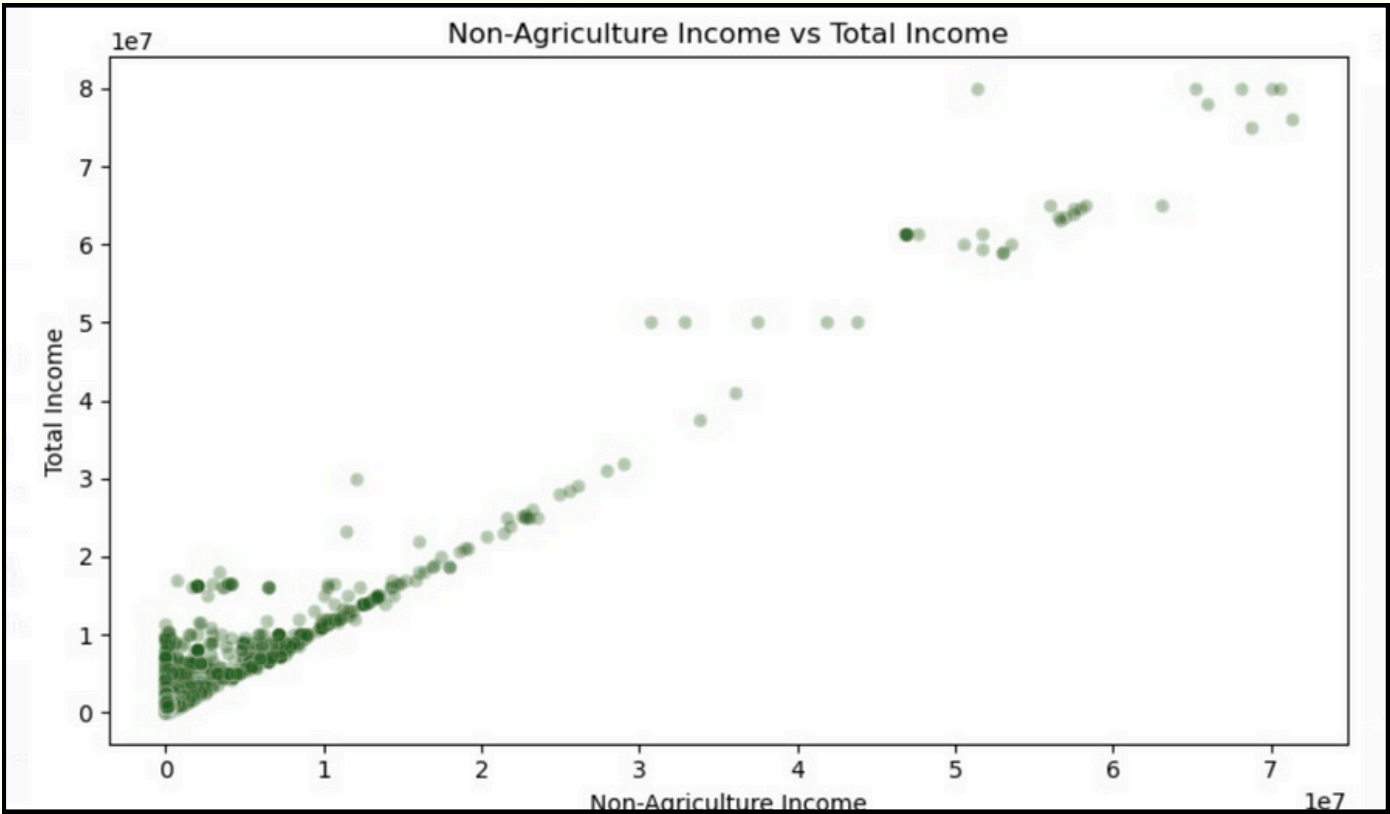
- Addressed missing values in key features (e.g., land size, ownership)
- Applied KNN imputation for different data to handle skew
- Cleaned and encoded categorical variables (e.g., "missing_info" as placeholder)

- The farmer's income is highly right-skewed.
- Majority of farmers fall in the lower income range, with a small segment earning disproportionately more.
- This distribution highlights significant income disparity across the population.

- Income is also influenced by how intensively the land is used, not just size.
- Two high-impact predictors: 'K022-Net Agri area and Land Holding Index source
- These suggest that efficient land use and land per capita both drive income.

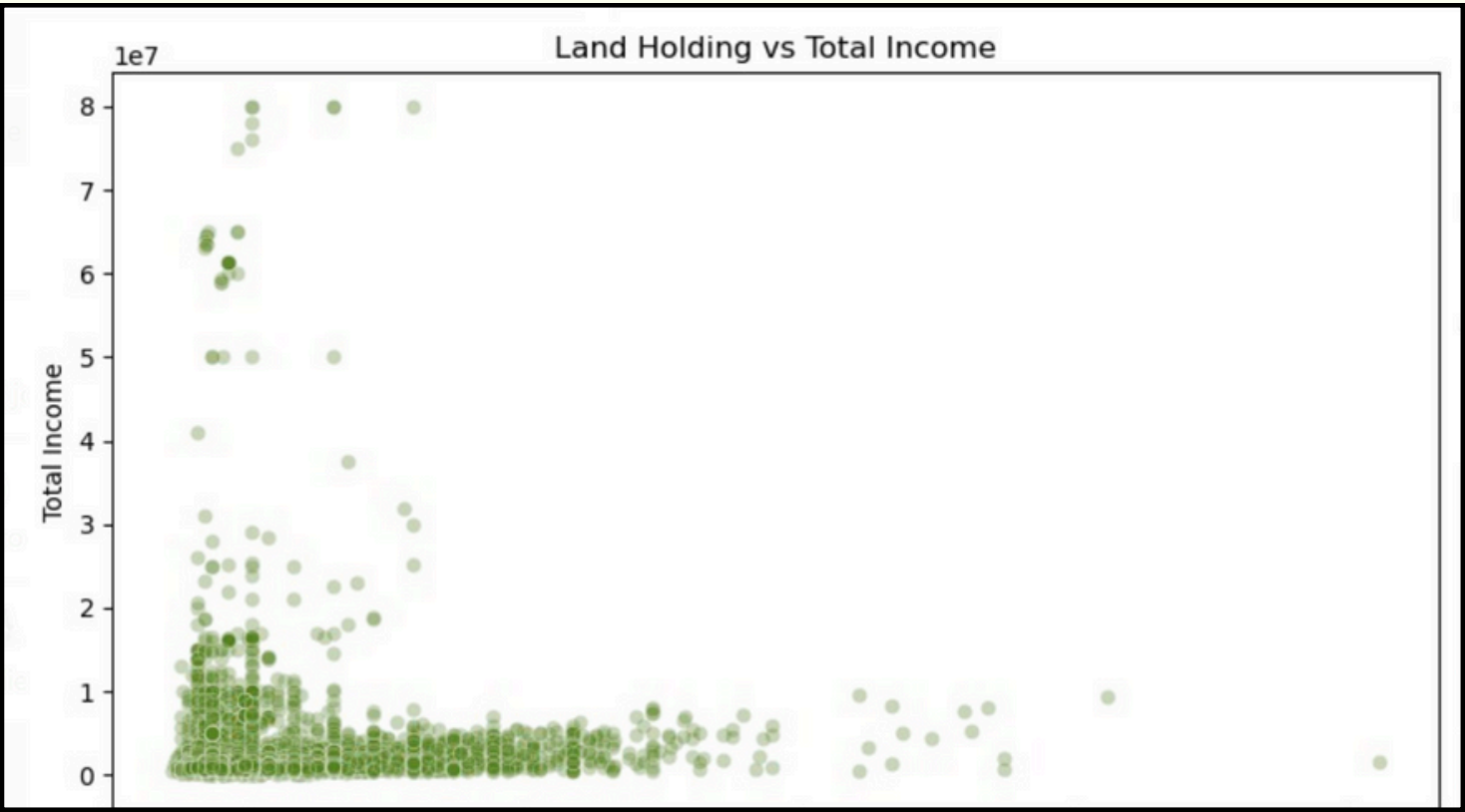
- Strong positive correlation between:
'Total_Land_For_Agriculture' and 'K022-net Agri area'
- Farmers with larger land holdings and greater net cultivated area tend to have higher incomes.





Non-Agriculture Income vs Total Income:

There is a strong positive correlation between non-agriculture income and total income. This indicates that diversification beyond farming significantly boosts household earnings.



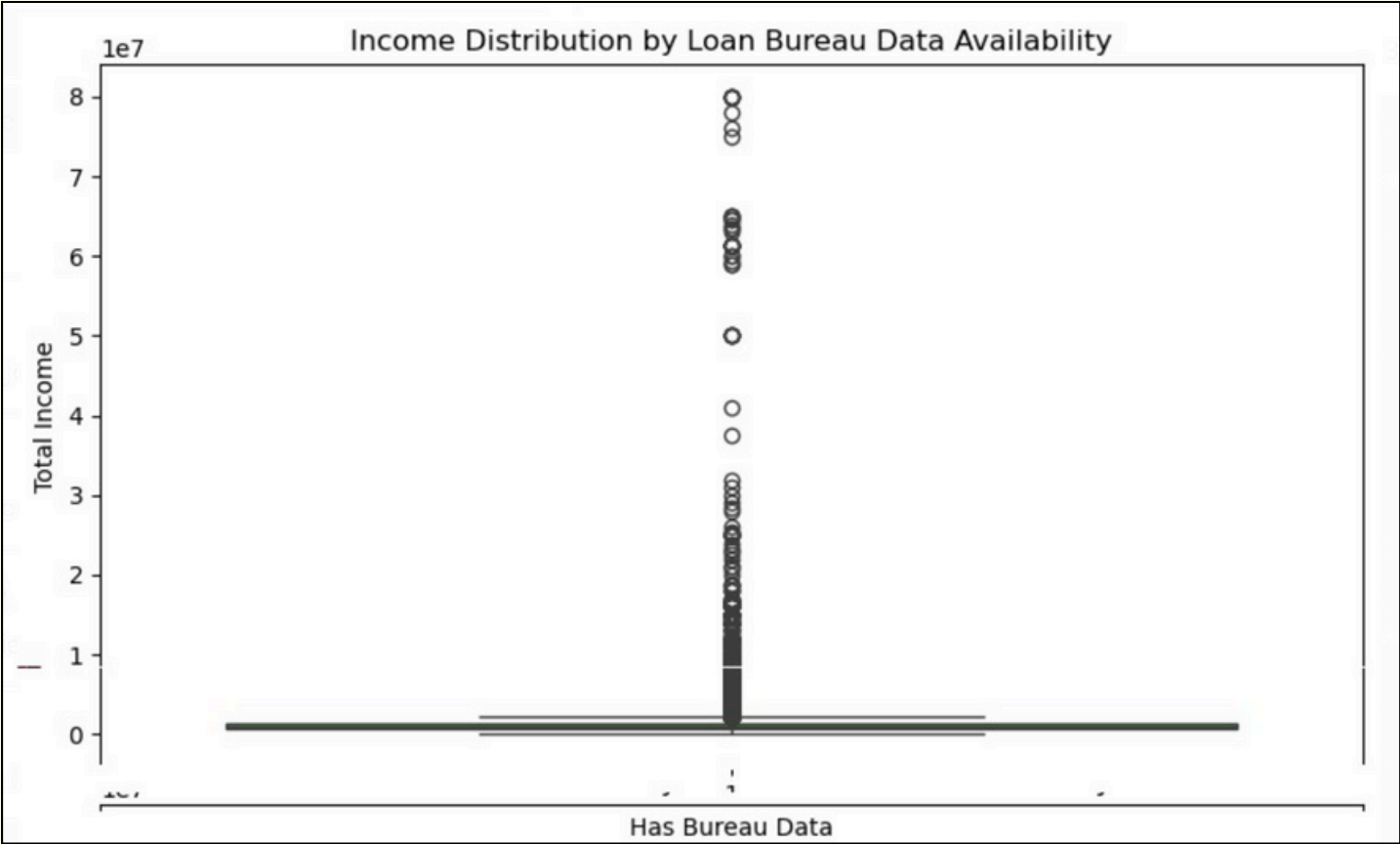
Land Holding vs Total Income:

Surprisingly, larger land holdings do not consistently translate to higher income. Many high-income farmers have small land areas, indicating the role of alternative income sources.



Housing Quality vs Total Income:

No clear relationship is observed between housing quality scores and total income. This suggests that housing conditions may not directly reflect income levels in the dataset.



Loan Bureau Data Availability vs Total Income:

Farmers with loan bureau data show high income variability, with some extreme outliers. This could indicate greater access to credit or more formal financial engagement.

GRAPHS

Data Preparation: Engineering Better Features



Handling Missing Data

Direct Imputation:

Replaced missing values in avg_disbursement_amount_bureau with 0, assuming that no data implies no formal disbursement.

KNN Imputation Setup:

Selected ~20+ numeric features such as rainfall, irrigation, and land area to be used in KNN Imputation, a technique that fills missing values by finding the most similar entries in the dataset.

Binary Flags:

Created logical indicators for presence/absence of data (e.g., has_loan_bureau_data) as an informative feature for the model.



Data Transformation

- Missing Value Handling: Custom imputation was applied for both numerical and categorical features.
- Categorical Encoding:
 - Label Encoding for low-cardinality features (e.g., sex, region).
 - Target Encoding with cross-validation for high-cardinality features (e.g., district, mandi).
- Text Columns Transformation:
 - Water body type column split into multiple binary flags (e.g., near_river, near_wetland).
 - Temperature columns split into min/max values.
- Duplicate & Redundant Columns: Removed after checking for similarity and correlation.
- Column Cleanup: Dropped original categorical columns after encoding to avoid data leakage.



Feature Engineering

- To extract deeper insights and relationships, the following new features were created:
- Relative Income Flags:
 - Added binary indicators showing whether a farmer's income is above the average income in their region, state, and district.
- Domain-Specific Features:
 - land_per_active_loan: Total agricultural land per active loan.
 - agri_income_ratio: Ratio of total income to non-agricultural income.
 - avg_rainfall: Mean rainfall over multiple seasonal records.
 - housing_quality_score: Averaged score of housing indicators like room size, electricity, and housing quality.
 - groundwater_variability: Standard deviation across groundwater thickness features.
 - market_access_score & rail_access_score: Inverse proximity to nearest mandi/railway station.

Building a Predictive Model

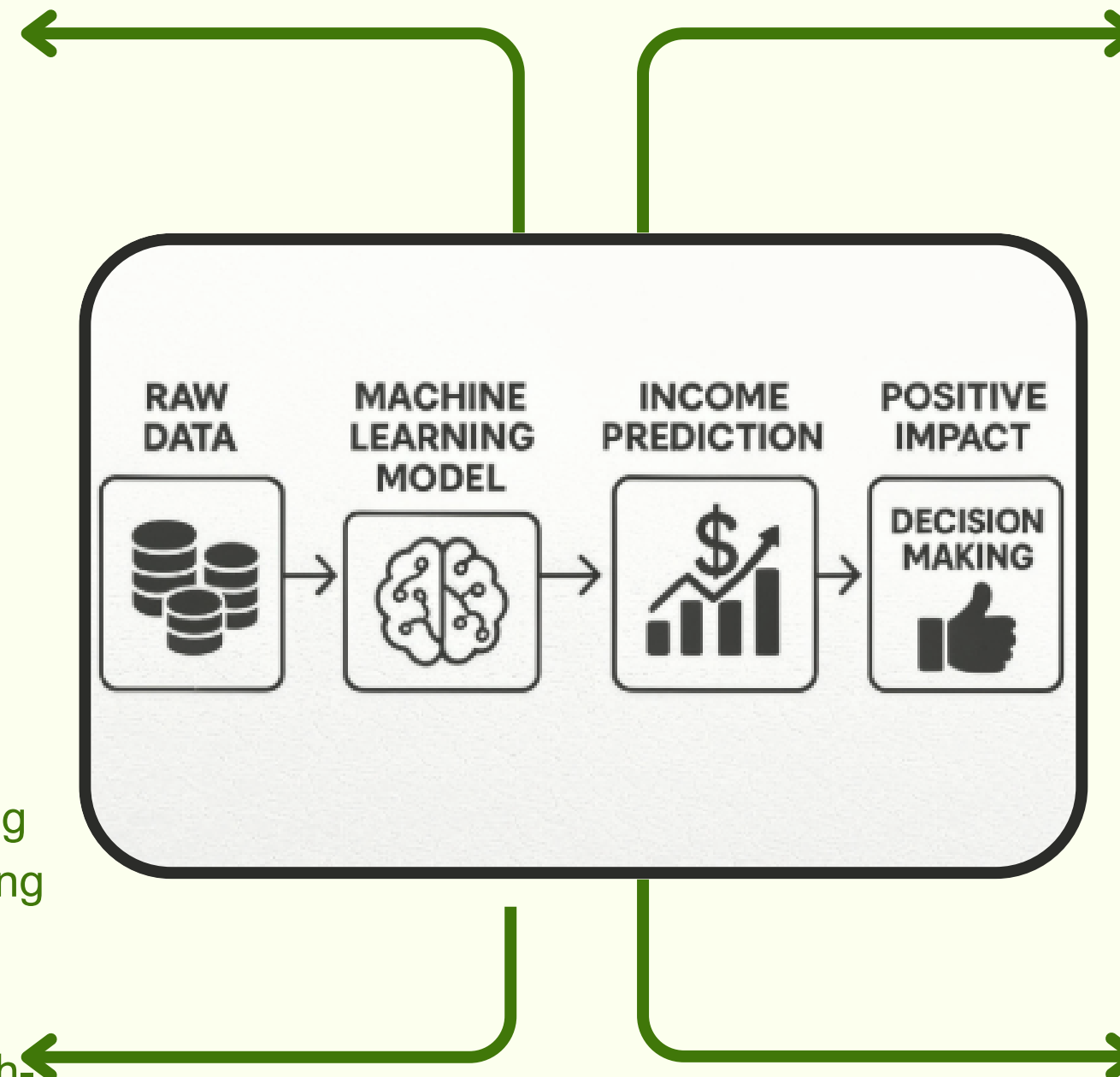
Core Goal: To develop a robust ML model that accurately predicts farmers' total income from agricultural and non-agricultural sources.

Why XGBoost?

- Performs exceptionally well on structured/tabular data.
- Handles missing values internally and prevents overfitting via regularization.
- Captures complex non-linear patterns in farmer income.
- Provides feature importance and works well with engineered features

Training Process:

- **Model:** XGBoost Regressor with default parameters.
- **Evaluation:** 5-fold Cross-Validation using Mean Absolute Error (MAE) as the scoring metric.
- **Features:** Included label-encoded low-cardinality variables, target-encoded high-cardinality variables, numerical data, and domain-inspired engineered features.



Why Not Other Models?

- Linear Regression: Too simplistic for complex relationships in income data.
- CatBoost/LightGBM: Similar to XGBoost, but XGBoost provided better performance in trials so we didn't use it in the final model.
- Neural Networks: Require more data and tuning; less interpretable for tabular data.
- Ensemble Models: Not used to keep the pipeline simple and interpretable.

Performance:

- Achieved an average cross-validated MAE of approximately ₹1.5 lakhs.
- This represents ~12–13% of the average farmer income, indicating reasonably accurate performance in a challenging real-world dataset.

Cross-Validation & Model Performance Evaluation

SEGMENT-WISE PERFORMANCE (POST-TUNING)

Low Income

MAE: ₹10,023.27
RMSE: ₹23,766.55
R²: 0.9732

Mid Income

MAE: ₹27,709.40
RMSE: ₹70,814.40
R²: 0.9797

High Income

MAE: ₹817,038.50
RMSE: ₹1,857,957.00
R²: 0.9452

Why Cross-Validation?

- Ensures model generalization by testing on unseen folds of data.
- Reduces the risk of overfitting or underfitting.
- Used 3-fold cross-validation to balance compute time and robustness.
- Helps in preventing data leakage.

Performance Metrics Used

- MAE (Mean Absolute Error): Measures average prediction error in ₹.
- RMSE (Root Mean Squared Error): Penalizes large errors more heavily.
- R² Score: Indicates how well the model explains variance in income data.

Comparative Model Assessment

- LightGBM showed strongest performance for Low & Mid segments.
- XGBoost outperformed others for High Income due to handling of extreme values.
- Final ensemble model chosen based on best segment-wise performance.

Final Conclusion

SUMMARY & MODELING ROADMAP AHEAD

Summary of Findings:

- Income prediction accuracy improved significantly through segmentation (Low, Mid, High income).
- Ensemble of XGBoost yielded the best performance with optimized hyperparameters.
- R^2 scores of 0.95+ across all income brackets demonstrate strong model generalization.
- LightGBM showed highest accuracy for Low Income, while XGBoost performed better for High Income after fine-tuning.
- Log transformation and feature scaling helped stabilize variance and improve prediction quality.
- Mixed-model strategy proved crucial in addressing income variability across segments.

IDEAS FOR FUTURE MODELING:

- Explore stacked ensemble methods using meta-learners to learn from base model outputs.
- Incorporate more geospatial and temporal features if available (e.g., crop cycle timing, climate data).
- Apply SHAP values or permutation importance for deeper interpretability and feature impact understanding.
- Fine-tune CatBoost with class-specific objective weights for better performance in skewed segments.

Key Challenges Faced:

- High skewness in income distribution created difficulty in building a single global model.
- Outliers in high-income group disproportionately impacted RMSE and MAE.
- Feature inconsistency between training and test datasets required careful pipeline design.
- Missing or noisy categorical data required robust encoding and careful treatment.
- Model blending and inference orchestration for segmented predictions demanded customized logic.
- Training models on smaller high-income group led to overfitting risks and limited generalization.

Thank You!

